

LES ACL:

Routeur R1.3:

SITEP_FILTER_IN_VLAN10

```
ip access-list extended SITEP_FILTER_IN_VLAN10
permit ospf any any
permit udp 192.168.10.0 0.0.0.31 host 192.168.0.4 eq 53
permit tcp 192.168.10.0 0.0.0.31 host 192.168.0.4 eq 53
permit udp 192.168.10.0 0.0.0.31 host 192.168.0.3 eq 67
permit udp 192.168.10.0 0.0.0.31 host 192.168.0.3 eq 68
permit tcp 192.168.10.0 0.0.0.31 host 192.168.0.5 eq 137
permit tcp 192.168.10.0 0.0.0.31 host 192.168.0.5 eq 138
permit tcp 192.168.10.0 0.0.0.31 host 192.168.0.5 eq 139
permit tcp 192.168.10.0 0.0.0.31 host 192.168.0.5 eq 145
permit tcp 192.168.10.0 0.0.0.31 host 192.168.0.7 eq 143
permit tcp 192.168.10.0 0.0.0.31 host 192.168.0.8 eq 587
permit tcp 192.168.10.0 0.0.0.31 host 192.168.0.6 eq 80
permit tcp 192.168.10.0 0.0.0.31 host 192.168.0.6 eq 443
```

SITEP_FILTER_OUT_VLAN10

```
ip access-list extended SITEP_FILTER_OUT_VLAN10
permit ospf any any
permit udp host 192.168.0.4 eq 53 192.168.10.0 0.0.0.31
permit tcp host 192.168.0.4 eq 53 192.168.10.0 0.0.0.31
permit udp host 192.168.0.3 eq 67 192.168.10.0 0.0.0.31
```

```
permit udp host 192.168.0.3 eq 68 192.168.10.0 0.0.0.31
permit tcp host 192.168.0.5 eq 137 192.168.10.0 0.0.0.31
permit tcp host 192.168.0.5 eq 138 192.168.10.0 0.0.0.31
permit tcp host 192.168.0.5 eq 139 192.168.10.0 0.0.0.31
permit tcp host 192.168.0.5 eq 145 192.168.10.0 0.0.0.31
permit tcp host 192.168.0.7 eq 143 192.168.10.0 0.0.0.31
permit tcp host 192.168.0.8 eq 587 192.168.10.0 0.0.0.31
permit tcp host 192.168.0.6 eq 80 192.168.10.0 0.0.0.31
permit tcp host 192.168.0.6 eq 443 192.168.10.0 0.0.0.31
```

SITEP_FILTER_IN_VLAN20

```
ip access-list extended SITEP_FILTER_IN_VLAN20
permit ospf any any
permit udp 192.168.20.0 0.0.0.63 host 192.168.0.4 eq 53
permit tcp 192.168.20.0 0.0.0.63 host 192.168.0.4 eq 53
permit udp 192.168.20.0 0.0.0.63 host 192.168.0.3 eq 67
permit udp 192.168.20.0 0.0.0.63 host 192.168.0.3 eq 68
permit tcp 192.168.20.0 0.0.0.63 host 192.168.0.5 eq 137
permit tcp 192.168.20.0 0.0.0.63 host 192.168.0.5 eq 138
permit tcp 192.168.20.0 0.0.0.63 host 192.168.0.5 eq 139
permit tcp 192.168.20.0 0.0.0.63 host 192.168.0.5 eq 145
permit tcp 192.168.20.0 0.0.0.63 host 192.168.0.7 eq 143
permit tcp 192.168.20.0 0.0.0.63 host 192.168.0.8 eq 587
permit tcp 192.168.20.0 0.0.0.63 host 192.168.0.6 eq 80
permit tcp 192.168.20.0 0.0.0.63 host 192.168.0.6 eq 443
```

SITEP_FILTER_OUT_VLAN20

```
ip access-list extended SITEP_FILTER_OUT_VLAN20
permit ospf any any
permit udp host 192.168.0.4 eq 53 192.168.20.0 0.0.0.63
permit tcp host 192.168.0.4 eq 53 192.168.20.0 0.0.0.63
permit udp host 192.168.0.3 eq 67 192.168.20.0 0.0.0.63
permit udp host 192.168.0.3 eq 68 192.168.20.0 0.0.0.63
permit tcp host 192.168.0.5 eq 137 192.168.20.0 0.0.0.63
permit tcp host 192.168.0.5 eq 138 192.168.20.0 0.0.0.63
permit tcp host 192.168.0.5 eq 139 192.168.20.0 0.0.0.63
permit tcp host 192.168.0.5 eq 145 192.168.20.0 0.0.0.63
permit tcp host 192.168.0.7 eq 143 192.168.20.0 0.0.0.63
permit tcp host 192.168.0.8 eq 587 192.168.20.0 0.0.0.63
permit tcp host 192.168.0.6 eq 80 192.168.20.0 0.0.0.63
permit tcp host 192.168.0.6 eq 443 192.168.20.0 0.0.0.63
```

Routeur R2.4:

SITEP_FILTER_IN_VLAN30

```
ip access-list extended SITEP_FILTER_IN_VLAN30
permit ospf any any
permit udp 192.168.30.0 0.0.0.31 host 192.168.0.4 eq 53
permit tcp 192.168.30.0 0.0.0.31 host 192.168.0.4 eq 53
permit udp 192.168.30.0 0.0.0.31 host 192.168.0.3 eq 67
permit udp 192.168.30.0 0.0.0.31 host 192.168.0.3 eq 68
permit tcp 192.168.30.0 0.0.0.31 host 192.168.0.5 eq 137
permit tcp 192.168.30.0 0.0.0.31 host 192.168.0.5 eq 138
```

```
permit tcp 192.168.30.0 0.0.0.31 host 192.168.0.5 eq 139
permit tcp 192.168.30.0 0.0.0.31 host 192.168.0.5 eq 145
permit tcp 192.168.30.0 0.0.0.31 host 192.168.0.7 eq 143
permit tcp 192.168.30.0 0.0.0.31 host 192.168.0.8 eq 587
permit tcp 192.168.30.0 0.0.0.31 host 192.168.0.6 eq 80
permit tcp 192.168.30.0 0.0.0.31 host 192.168.0.6 eq 443
```

SITEP_FILTER_OUT_VLAN30

```
ip access-list extended SITEP_FILTER_OUT_VLAN30
permit ospf any any
permit udp host 192.168.0.4 eq 53 192.168.30.0 0.0.0.31
permit tcp host 192.168.0.4 eq 53 192.168.30.0 0.0.0.31
permit udp host 192.168.0.3 eq 67 192.168.30.0 0.0.0.31
permit udp host 192.168.0.3 eq 68 192.168.30.0 0.0.0.31
permit tcp host 192.168.0.5 eq 137 192.168.30.0 0.0.0.31
permit tcp host 192.168.0.5 eq 138 192.168.30.0 0.0.0.31
permit tcp host 192.168.0.5 eq 139 192.168.30.0 0.0.0.31
permit tcp host 192.168.0.5 eq 145 192.168.30.0 0.0.0.31
permit tcp host 192.168.0.7 eq 143 192.168.30.0 0.0.0.31
permit tcp host 192.168.0.8 eq 587 192.168.30.0 0.0.0.31
permit tcp host 192.168.0.6 eq 80 192.168.30.0 0.0.0.31
permit tcp host 192.168.0.6 eq 443 192.168.30.0 0.0.0.31
```

SITEP_FILTER_IN_VLAN40

```
ip access-list extended SITEP_FILTER_IN_VLAN40
permit ospf any any
permit udp 192.168.40.0 0.0.0.127 host 192.168.0.4 eq 53
permit tcp 192.168.40.0 0.0.0.127 host 192.168.0.4 eq 53
permit udp 192.168.40.0 0.0.0.127 host 192.168.0.3 eq 67
permit udp 192.168.40.0 0.0.0.127 host 192.168.0.3 eq 68
permit tcp 192.168.40.0 0.0.0.127 host 192.168.0.5 eq 137
permit tcp 192.168.40.0 0.0.0.127 host 192.168.0.5 eq 138
permit tcp 192.168.40.0 0.0.0.127 host 192.168.0.5 eq 139
permit tcp 192.168.40.0 0.0.0.127 host 192.168.0.5 eq 145
permit tcp 192.168.40.0 0.0.0.127 host 192.168.0.7 eq 143
permit tcp 192.168.40.0 0.0.0.127 host 192.168.0.8 eq 587
permit tcp 192.168.40.0 0.0.0.127 host 192.168.0.6 eq 80
permit tcp 192.168.40.0 0.0.0.127 host 192.168.0.6 eq 443
```

SITEP_FILTER_OUT_VLAN40

```
ip access-list extended SITEP_FILTER_OUT_VLAN40
permit ospf any any
permit udp host 192.168.0.4 eq 53 192.168.40.0 0.0.0.127
permit tcp host 192.168.0.4 eq 53 192.168.40.0 0.0.0.127
permit udp host 192.168.0.3 eq 67 192.168.40.0 0.0.0.127
permit udp host 192.168.0.3 eq 68 192.168.40.0 0.0.0.127
permit tcp host 192.168.0.5 eq 137 192.168.40.0 0.0.0.127
permit tcp host 192.168.0.5 eq 138 192.168.40.0 0.0.0.127
```

```
permit tcp host 192.168.0.5 eq 139 192.168.40.0 0.0.0.127
permit tcp host 192.168.0.5 eq 145 192.168.40.0 0.0.0.127
permit tcp host 192.168.0.7 eq 143 192.168.40.0 0.0.0.127
permit tcp host 192.168.0.8 eq 587 192.168.40.0 0.0.0.127
permit tcp host 192.168.0.6 eq 80 192.168.40.0 0.0.0.127
permit tcp host 192.168.0.6 eq 443 192.168.40.0 0.0.0.127
```

Routeur R2.6:

SITEP_FILTER_IN_VLAN50

```
ip access-list extended SITEP_FILTER_IN_VLAN50
permit ospf any any
permit udp 192.168.50.0 0.0.0.15 host 192.168.0.4 eq 53
permit tcp 192.168.50.0 0.0.0.15 host 192.168.0.4 eq 53
permit udp 192.168.50.0 0.0.0.15 host 192.168.0.3 eq 67
permit udp 192.168.50.0 0.0.0.15 host 192.168.0.3 eq 68
permit tcp 192.168.50.0 0.0.0.15 host 192.168.0.5 eq 137
permit tcp 192.168.50.0 0.0.0.15 host 192.168.0.5 eq 138
permit tcp 192.168.50.0 0.0.0.15 host 192.168.0.5 eq 139
permit tcp 192.168.50.0 0.0.0.15 host 192.168.0.5 eq 145
permit tcp 192.168.50.0 0.0.0.15 host 192.168.0.7 eq 143
permit tcp 192.168.50.0 0.0.0.15 host 192.168.0.8 eq 587
permit tcp 192.168.50.0 0.0.0.15 host 192.168.0.6 eq 80
permit tcp 192.168.50.0 0.0.0.15 host 192.168.0.6 eq 443
```

SITEP_FILTER_OUT_VLAN50

```
ip access-list extended SITEP_FILTER_OUT_VLAN50
permit ospf any any
permit udp host 192.168.0.4 eq 53 192.168.50.0 0.0.0.15
permit tcp host 192.168.0.4 eq 53 192.168.50.0 0.0.0.15
permit udp host 192.168.0.3 eq 67 192.168.50.0 0.0.0.15
permit udp host 192.168.0.3 eq 68 192.168.50.0 0.0.0.15
permit tcp host 192.168.0.5 eq 137 192.168.50.0 0.0.0.15
permit tcp host 192.168.0.5 eq 138 192.168.50.0 0.0.0.15
permit tcp host 192.168.0.5 eq 139 192.168.50.0 0.0.0.15
permit tcp host 192.168.0.5 eq 145 192.168.50.0 0.0.0.15
permit tcp host 192.168.0.7 eq 143 192.168.50.0 0.0.0.15
permit tcp host 192.168.0.8 eq 587 192.168.50.0 0.0.0.15
permit tcp host 192.168.0.6 eq 80 192.168.50.0 0.0.0.15
permit tcp host 192.168.0.6 eq 443 192.168.50.0 0.0.0.15
```

SITEP_FILTER_IN_VLAN60

```
ip access-list extended SITEP_FILTER_IN_VLAN60
permit ospf any any
permit udp 192.168.60.0 0.0.0.31 host 192.168.0.4 eq 53
permit tcp 192.168.60.0 0.0.0.31 host 192.168.0.4 eq 53
permit udp 192.168.60.0 0.0.0.31 host 192.168.0.3 eq 67
permit udp 192.168.60.0 0.0.0.31 host 192.168.0.3 eq 68
permit tcp 192.168.60.0 0.0.0.31 host 192.168.0.5 eq 137
permit tcp 192.168.60.0 0.0.0.31 host 192.168.0.5 eq 138
permit tcp 192.168.60.0 0.0.0.31 host 192.168.0.5 eq 139
```

```
permit tcp 192.168.60.0 0.0.0.31 host 192.168.0.5 eq 145
permit tcp 192.168.60.0 0.0.0.31 host 192.168.0.7 eq 143
permit tcp 192.168.60.0 0.0.0.31 host 192.168.0.8 eq 587
permit tcp 192.168.60.0 0.0.0.31 host 192.168.0.6 eq 80
permit tcp 192.168.60.0 0.0.0.31 host 192.168.0.6 eq 443
```

SITEP_FILTER_OUT_VLAN60

```
ip access-list extended SITEP_FILTER_OUT_VLAN60
permit ospf any any
permit udp host 192.168.0.4 eq 53 192.168.60.0 0.0.0.31
permit tcp host 192.168.0.4 eq 53 192.168.60.0 0.0.0.31
permit udp host 192.168.0.3 eq 67 192.168.60.0 0.0.0.31
permit udp host 192.168.0.3 eq 68 192.168.60.0 0.0.0.31
permit tcp host 192.168.0.5 eq 137 192.168.60.0 0.0.0.31
permit tcp host 192.168.0.5 eq 138 192.168.60.0 0.0.0.31
permit tcp host 192.168.0.5 eq 139 192.168.60.0 0.0.0.31
permit tcp host 192.168.0.5 eq 145 192.168.60.0 0.0.0.31
permit tcp host 192.168.0.7 eq 143 192.168.60.0 0.0.0.31
permit tcp host 192.168.0.8 eq 587 192.168.60.0 0.0.0.31
permit tcp host 192.168.0.6 eq 80 192.168.60.0 0.0.0.31
permit tcp host 192.168.0.6 eq 443 192.168.60.0 0.0.0.31
```


Tunnel VPN ipsec:

Routeur R2.5

```
crypto isakmp policy 10
  encr aes
  authentication pre-share
  group 2
  exit

crypto isakmp key CLE_SECRETE address 20.0.0.2

crypto ipsec transform-set TS-R2-SET esp-aes esp-sha-hmac
  mode tunnel

ip access-list extended CRYPTO-R2.5-TO-R2.6
  permit ip 192.168.0.0 0.0.0.15 192.168.5.0 0.0.0.15
  permit ip 192.168.0.0 0.0.0.15 192.168.6.0 0.0.0.31

crypto map CMAP-R2 10 ipsec-isakmp
  set peer 20.0.0.2
  set transform-set TS-R2-SET
  match address CRYPTO-R2.5-TO-R2.6

interface GigabitEthernet0/0
  crypto map CMAP-R2
```

Routeur R2.6

```
crypto isakmp policy 10
  encr aes
  authentication pre-share
  group 2
  exit

crypto isakmp key CLE_SECRETE address 20.0.0.1

crypto ipsec transform-set TS-R2-SET esp-aes esp-sha-hmac
  mode tunnel

ip access-list extended CRYPTO-R2.6-TO-R2.5
  permit ip 192.168.5.0 0.0.0.15 192.168.0.0 0.0.0.15
  permit ip 192.168.6.0 0.0.0.31 192.168.0.0 0.0.0.15

crypto map CMAP-R2 10 ipsec-isakmp
  set peer 20.0.0.1
  set transform-set TS-R2-SET
  match address CRYPTO-R2.6-TO-R2.5

interface GigabitEthernet0/0
  crypto map CMAP-R2
```